

Abhigna Kandala

Software Developer | Full-Stack and Cloud Engineering | React, TypeScript, Python, Node.js, AWS
College Park, MD | +1 2407918390 | k.abhigna2@gmail.com | [LinkedIn](#) | github.com/KandalaAbhigna

Professional Summary

M.Eng. Software Engineering student (4.00 GPA) who co-built a live stock-intelligence platform on Amazon Web Services (AWS) as one of two engineers: an automated ECS Fargate and Step Functions pipeline scoring about 4,500 tickers across 5 market universes, a Flask REST API and React/TypeScript dashboard served through Lambda, and a 4-intern program she supervised and code-reviewed. Also shipped a campus event platform (FastAPI, React, 27 endpoints, Kubernetes CI/CD, 200+ automated tests) and a MERN-stack app on EC2. Strong in Python, TypeScript, React, REST API design, AWS, Docker, Kubernetes, and CI/CD.

Skills

- **Languages and Frameworks:** Python, Java, C++, TypeScript, JavaScript, SQL, HTML, CSS, React, Node.js, Express.js, FastAPI, Flask
- **Backend and APIs:** REST API design, microservices, JSON Web Token (JWT) authentication, role-based access control
- **Cloud and DevOps:** AWS (EC2, ECS Fargate, Lambda, S3, RDS, Step Functions, EventBridge, ECR, IAM), Docker, Kubernetes (ingress, autoscaling, cert-manager), GitHub Actions, Continuous Integration/Continuous Deployment (CI/CD), Linux, Git
- **Databases and Data:** MySQL, PostgreSQL, MongoDB, schema design, Pandas, NumPy, Scikit-Learn, LightGBM, PyTorch, TensorFlow
- **Testing and Practices:** Pytest, Vitest, code review, technical documentation, Agile/Scrum, SDLC, Object-Oriented Design

PROFESSIONAL EXPERIENCE

Firstsource Solutions Ltd. | *Data Analyst Intern*

May 2024 - Dec 2024

- Developed Python scripts automating recurring data-processing workflows for a cross-functional team of analysts and machine learning (ML) engineers, cutting manual effort and improving the reliability of scheduled jobs.
- Led code reviews using GitHub pull requests, providing specific, actionable feedback that aligned with team SDLC standards, and authored clear technical documentation, which improved code consistency and shortened review cycles

Savart | *Software Development Engineer (SDE) Intern*

Jun 2023 - Nov 2023

- Built a complete internal research web page from scratch, designing the layout, components, styling, form handling, and client-side state using JavaScript, HTML, and CSS, which enabled researchers to submit queries without external tools
- Built RESTful APIs using TypeScript and Node.js, integrated the page with backend MySQL services, and wrote clean, maintainable code following team conventions, which reduced API response errors and streamlined data retrieval for researchers
- Delivered features on schedule in Agile sprints, participating in sprint planning and daily standups with a distributed team.

Skolar | *Software Engineering Intern*

Oct 2022 - Dec 2022

- Implemented core data structures (arrays, linked lists, stacks, queues, hash tables, trees, heaps, graphs) and their algorithms in Python on Linux, then integrated them into the project codebase, enabling more efficient data handling
- Developed and documented TypeScript code, wrote unit tests, and followed SDLC guidelines while using Docker containers and conducting code reviews, which reduced bugs and improved code reliability

PROJECTS

Siddhantha T20: Cloud-Native Stock Prediction Platform

Jan 2026 - Jun 2026

- Python, Flask, React, TypeScript, LightGBM, AWS
- Co-built, as one of two engineers reporting to the tech lead, a platform that scores about 4,500 tickers across 5 market universes with LightGBM models and serves daily Top-20 stock picks to a live web dashboard; automated the end-to-end pipeline (download, dataset build, features, prediction, stats) on AWS ECS Fargate, Step Functions, and EventBridge, replacing manual daily runs.
- Developed the Flask REST API over Amazon S3 datasets and RDS MySQL, served through Lambda, and the React/TypeScript dashboard it powers, with fault-tolerant error handling and optimized data fetching; extended coverage from 3 to 5 universes and added the Russell 1000 (135,000+ feature rows, 451 tickers), fixing two latent production bugs.
- Backtested a proposed swing-trading strategy and delivered a clear no-go recommendation (profit factor 0.84, 32.7% win rate) that the tech lead adopted, stopping build-out before more engineering time was spent; supervised and code-reviewed 4 software interns, producing a prioritized 10-item review and a strict-TypeScript reference implementation with handover docs.

Terrapin Events: Campus Event Management System

- React, TypeScript, FastAPI, MongoDB, Docker, Kubernetes, GitHub Actions
- Built, in a graduate software design team, a full-stack platform where organizers submit events for admin approval, participants register with an automated waitlist, and the system emails confirmations, QR-code tickets, and feedback requests: a FastAPI backend with 27 REST endpoints and JWT role-based access control (Admin, Organizer, Participant) plus a 14-page React/TypeScript front end.
- Wrote 110 Vitest and 94 Pytest tests (87% backend coverage) and set up GitHub Actions CI/CD that builds and pushes Docker images and deploys to Kubernetes with Nginx ingress, cert-manager TLS, secrets, and pod autoscaling (2-10 replicas).

Digital Time Capsule: Multimedia Sharing Web App

- React.js, Node.js, Express.js, MongoDB, REST APIs, AWS EC2, Nginx
- Built a full-stack app for creating and sharing multimedia time capsules (photos, text, music) with friend requests, contributor permissions, and privacy controls: 25+ RESTful endpoints for authentication, capsule CRUD, friends, access control, and media storage, with JSON schema validation across 5 MongoDB collections.
- Built an 8-page React front end with reusable components and dynamic state management; deployed to production on EC2 behind Nginx as a reverse proxy, owning the full release path on Linux.

EDUCATION

University of Maryland, College Park | *M.Eng., Software Engineering* (GPA: 4.00 / 4.00)

Jan 2025 - Dec 2026

Kalinga Institute of Industrial Technology (KIIT) | *B.Tech, Computer Science and Engineering* (GPA: 8.81 / 10.00)

2020 - 2024

Certifications

- AWS Training and Certification
- HackerRank Problem Solving
- Full Stack Web Development, Edu-Versity

Achievements

- 250+ LeetCode solved